Run report: E4, two-torsion quotients of the S_3 system

2026-10-11, 17:14–17:15 UTC. research/isogeny_conductor_gap_round3_20261011, lead E4.

Command. `isogeny_gap_two_torsion_f4 --sizes 32,64 --budget-ms 20000` on the round-1 prime pairs (`research/isogeny_conductor_gap_ic_20261010/pairs/prime_pairs.json`). Runner `examples/isogeny_gap_two_torsion_f4.rs` (commit fb8185437 plus the orbit fix committed with the records); solver `cryptanalysis::f4_fp::f4`, grevlex, degree bound 2·deg + 8, 20 s budget. Host: cloud container, not an isolated ecbench host; wall times are observations only.

Workloads. 24 even-cofactor prime pairs (22 and 26 bits) × crater, floor, isomorphic floor model = 72 curves; base sizes B = 32, 64; subgroups 1 (direct), C2, C4, and E[2] where rational; two random and one planted target each.

Completed. 1,128 of 1,128 systems (752 random, 376 planted). No timeout; F4 agrees with enumeration on every system; every planted solution annihilates its basis; the Vélu map passed 32 random-sum checks per (curve, subgroup). A first launch looped on a C4-stable base whose orbit repeated an abscissa; it was stopped, fixed, and restarted from an empty record file.

Rational 2-torsion. Rank 2 on the crater and rank 1 on floor and model for all 12 D = −7 pairs and the D = −11 and D = −3 pairs with $\ell = 2$; rank 1 everywhere else.

Measured (exact counts; range over curves, roles, targets).

base-poly degree	solving degree	widest matrix	mults, random	mults, planted
8	10	45	7,408–7,414	13,903–13,929
16	18	93	34,152–34,158	81,128–81,178
32	34	189	145,630	483,552–483,561
64	66	381	600,510	3,123,348–3,123,354

Derived. Direct ÷ quotient at matched B: solving degree 1.89–3.67; multiplications 4.12–4.26 (|H| = 2) and 17.6–19.6 (|H| = 4). Crater E[2] against floor and model C4: same degree and matrix, multiplications within 6 of 34,158 at B = 64.

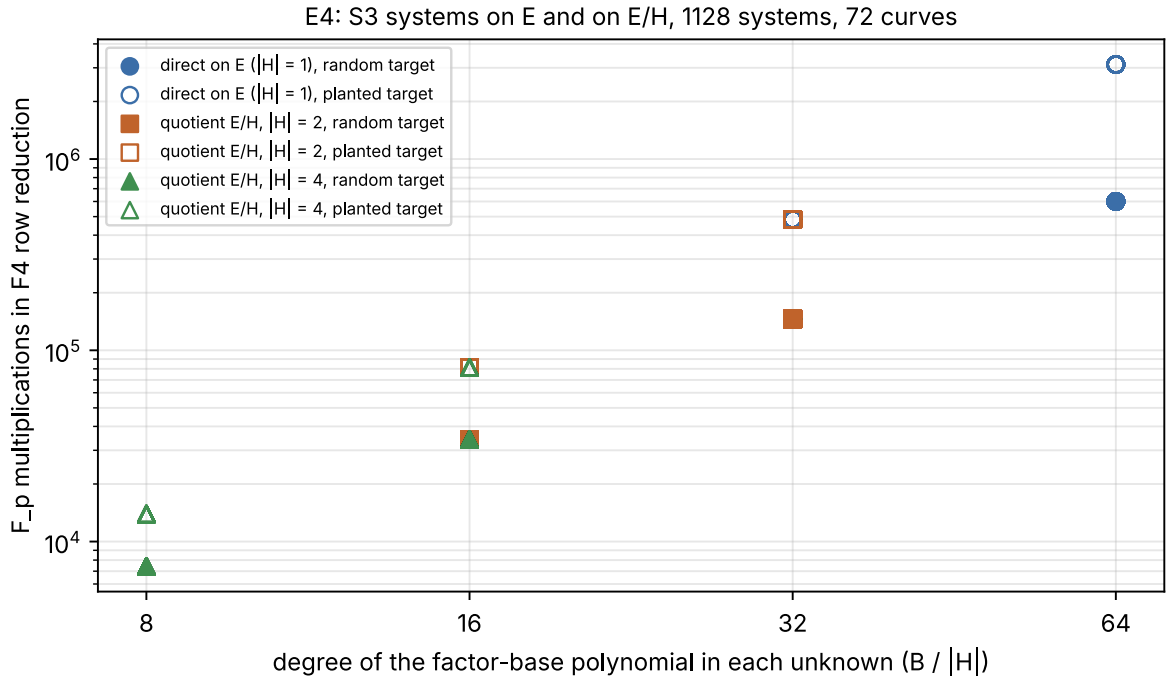


Figure 1: Every E4 system: F_p row-reduction multiplications against the factor-base polynomial degree B/|H|. 1,128 systems, 72 curves; records `e4/records.jsonl`.

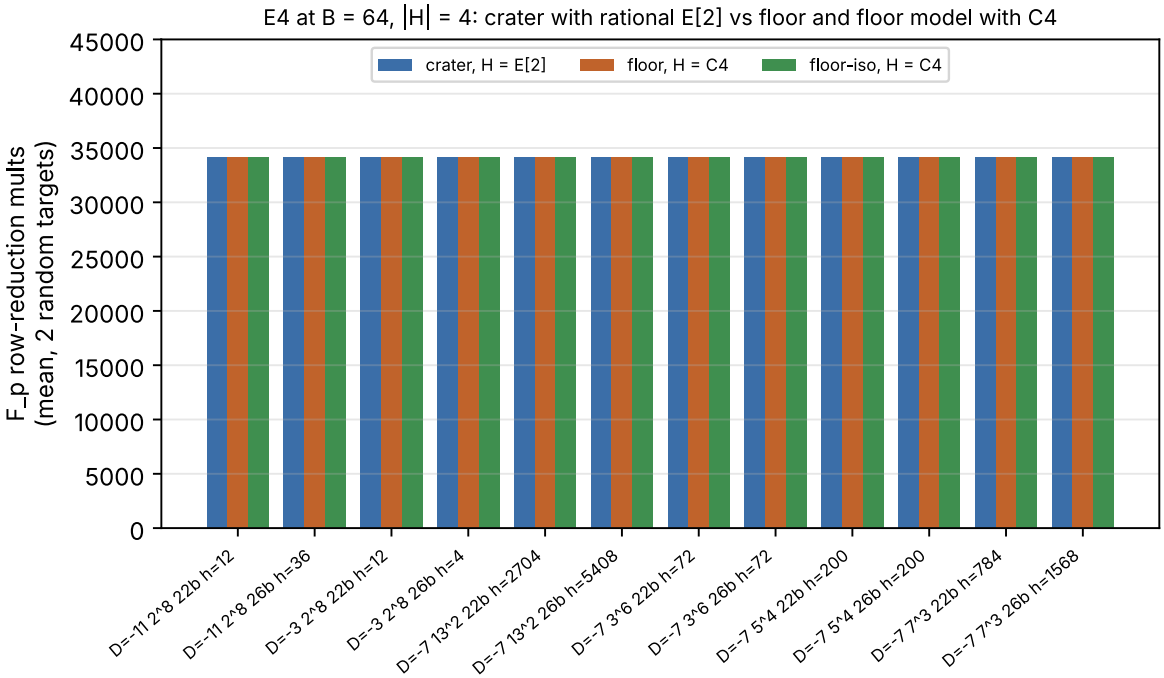


Figure 2: B = 64, |H| = 4: crater with rational E[2] against floor and floor model with C4, mean of 2 random targets per bar; the bars are equal to within 6 multiplications.

Raw records: `research/isogeny_conductor_gap_round3_20261011/e4/records.jsonl`; tables `TABLES.md`, `tables.json`; plot source `plot.py`.